A CLT for $\log|\det(I - U_N)|$

Theorem 1. *Let U_N be Haar distributed on $U(N)$, and fix $u \in \mathbb{C}$ with $|u| = 1$. Then*

$$\frac{\log|\det(I - U_N)|}{\sqrt{\frac{1}{2}\log N}} \xrightarrow[N\to\infty]{d} \mathcal{N}(0,1).$$

Proof. Put

$$X_N := \log|\det(I - U_N)|.$$

This random variable is well-defined almost surely, since the probability that 1 is an eigenvalue of U_N is zero. For $0 < r < 1$, define

$$X_N(r) := \log|\det(I - rU_N)|.$$

Inside the unit disk we have the absolutely convergent expansion

$$\log\det(I - rU_N) = -\sum_{k\geq 1} \frac{r^k}{k} \operatorname{Tr}(U_N^k).$$

Taking real parts gives

$$X_N(r) = -\sum_{k\geq 1} \frac{r^k}{k} \Re \operatorname{Tr}(U_N^k).$$

Write

$$T_{N,k} := \operatorname{Tr}(U_N^k).$$

We shall use the standard covariance identities (due to Dyson; proof omitted)

$$\mathbb{E}[T_{N,j}\overline{T_{N,k}}] = \delta_{jk}\min(j,N), \qquad \mathbb{E}[T_{N,j}T_{N,k}] = 0. \tag{1}$$

Therefore

$$\operatorname{Var} X_N(r) = \frac{1}{2}\sum_{k\geq 1} \frac{r^{2k}}{k^2}\min(k,N). \tag{2}$$

We set

$$r_N := e^{-1/N}.$$

We shall study $X_N(r_N)$. A direct computation shows that

$$\operatorname{Var} X_N(r_N) = \frac{1}{2}\sum_{k\geq 1} \frac{e^{-2k/N}}{k^2}\min(k,N).$$

Splitting the sum at $k = N$, we get

$$\sum_{k\leq N} \frac{e^{-2k/N}}{k} = \log N + O(1),$$

and

$$N\sum_{k>N} \frac{e^{-2k/N}}{k^2} = O(1).$$

Hence

$$\operatorname{Var} X_N(r_N) = \frac{1}{2}\log N + O(1). \tag{3}$$

We first prove a central limit theorem for $X_N(r_N)$. Let

$$M_N := \left\lfloor \frac{N}{\log N} \right\rfloor$$

and define the truncated sum

$$X_N^{(M)}(r_N) := -\sum_{k=1}^{M_N} \frac{r_N^k}{k} \Re T_{N,k}.$$

By (1),

$$\operatorname{Var}\big(X_N(r_N) - X_N^{(M)}(r_N)\big) = \frac{1}{2}\sum_{k>M_N} \frac{r_N^{2k}}{k^2}\min(k,N).$$

Since $r_N \le 1$,

$$\operatorname{Var}\big(X_N(r_N) - X_N^{(M)}(r_N)\big) \le \frac{1}{2}\sum_{M_N<k\le N}\frac{1}{k} + \frac{1}{2}N\sum_{k>N}\frac{1}{k^2} = O(\log\log N).$$

Consequently,

$$\frac{X_N(r_N) - X_N^{(M)}(r_N)}{\sqrt{\log N}} \longrightarrow 0 \tag{4}$$

in L^2, and hence in probability.

We now prove the central limit theorem for the truncated expression $X_N^{(M)}(r_N)$. We use the Diaconis–Shahshahani moment formula: if $a_1,\dots,a_m,b_1,\dots,b_m$ are nonnegative integers and $\sum_{j=1}^m ja_j = \sum_{j=1}^m jb_j \le N$, then

$$\mathbb{E}\prod_{j=1}^m T_{N,j}^{a_j}\overline{T_{N,j}}^{b_j} = \prod_{j=1}^m j^{a_j}a_j!\,\mathbf{1}_{\{a_j=b_j \text{ for all } j\}}. \tag{5}$$

If the two weighted sums are unequal, the expectation is zero by invariance under multiplication of U_N by a scalar on the unit circle.

Fix an integer $q \ge 1$. Since $M_N = \lfloor N/\log N\rfloor$, we have $qM_N \le N$ for all sufficiently large N. Thus (5) applies to every monomial appearing in the expansion of the first q moments of $X_N^{(M)}(r_N)$. It follows that these moments agree with the corresponding moments of

$$Z_N := -\sum_{k=1}^{M_N}\frac{r_N^k}{\sqrt{k}}\Re G_k,$$

where $G_1, G_2, \dots$ are independent standard complex Gaussian random variables. For a standard complex Gaussian we have that $\Re G_k$ is real Gaussian with variance $1/2$ and mean 0, so Z_N is a centered real Gaussian with variance

$$\operatorname{Var} Z_N = \frac{1}{2}\sum_{k=1}^{M_N}\frac{r_N^{2k}}{k}.$$

Since $r_N = e^{-1/N}$ and $M_N = N/\log N + O(1)$,

$$\sum_{k=1}^{M_N}\frac{r_N^{2k}}{k} = \sum_{k=1}^{M_N}\frac{1}{k} + O(1) = \log M_N + O(1) = \log N - \log\log N + O(1).$$

Thus

$$\operatorname{Var} Z_N = \frac{1}{2}\log N + o(\log N).$$

The preceding moment comparison implies that, for every fixed q, the q-th moment of

$$\frac{X_N^{(M)}(r_N)}{\sqrt{\frac{1}{2}\log N}}$$

converges to the q-th moment of $\mathcal{N}(0,1)$. Since the normal law is moment-determinate, the method of moments gives

$$\frac{X_N^{(M)}(r_N)}{\sqrt{\frac{1}{2}\log N}} \xrightarrow[N\to\infty]{d} \mathcal{N}(0,1). \tag{6}$$

Combining (4) and (6) by Slutsky's theorem yields

$$\frac{X_N(r_N)}{\sqrt{\frac{1}{2}\log N}} \xrightarrow[N\to\infty]{d} \mathcal{N}(0,1). \tag{7}$$

It remains to compare $X_N(r_N)$ with X_N. For $e^{i\theta} \in \mathbb{T}$, the Fourier series identities

$$\log|1-e^{i\theta}| = -\sum_{k\geq 1}\frac{\cos(k\theta)}{k}, \qquad \log|1-r_N e^{i\theta}| = -\sum_{k\geq 1}\frac{r_N^k\cos(k\theta)}{k}$$

hold in $L^2(\mathbb{T})$. Therefore

$$X_N - X_N(r_N) = -\sum_{k\geq 1}\frac{1-r_N^k}{k}\Re T_{N,k}$$

in L^2. Using (1),

$$\operatorname{Var}(X_N - X_N(r_N)) = \frac{1}{2}\sum_{k\geq 1}\frac{(1-r_N^k)^2}{k^2}\min(k,N).$$

For $k \leq N$, the bound $1-e^{-k/N} \leq k/N$ gives

$$\sum_{k\leq N}\frac{(1-r_N^k)^2}{k^2}k \leq \sum_{k\leq N}\frac{k}{N^2} = O(1).$$

For $k > N$,

$$N\sum_{k>N}\frac{(1-r_N^k)^2}{k^2} \leq N\sum_{k>N}\frac{1}{k^2} = O(1).$$

Hence

$$\operatorname{Var}(X_N - X_N(r_N)) = O(1),$$

and consequently

$$\frac{X_N - X_N(r_N)}{\sqrt{\log N}} \longrightarrow 0 \tag{8}$$

in L^2, hence in probability. The desired theorem now follows from (7), (8), and Slutsky's theorem. $\square$